

Xinru Cheng

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EDUCATION

Peking University

Beijing, China

Master of Software Engineering

Sep 2024 – Jul 2027

- Major Courses: Algorithm analysis and design, Game engines and game development, Natural language processing, Machine learning, Artificial intelligence practices, Software implementation technology, etc.
- Rank: 24/279 (Top 10%)

Beijing University of Chemical Technology

Beijing, China

Bachelor of Digital Media Art

Sep 2020 – Jul 2024

- Major Courses: Programming Foundation, Game Planning and Development, Advanced Mathematics, Advanced Practicum, Virtual Reality Technology, Three Dimensional Animation Design and Production, Graphic Image Processing, Practicum in Applied Software, Graphics of Engineering, etc.
- Rank: 1/30 (Direct Admission to Peking University)

RESEARCH INTERESTS

- Virtual, Augmented, and Mixed Reality Technologies
- Human-AI Interaction and Next-Generation Interfaces
- Human Behavior, Cognition, and Gamified Experiences in Interactive Systems

ACADEMIC PAPERS

- Zhang, Y., **Cheng, X.**, Yan, J. (2025).AI-Echo: exploring technological alienation through an interactive art experience. In *International Conference on Human-Computer Interaction (HCII 2025)* [C].
- Jiang, R., Zhang, S., Zhao, Y., **Cheng, X.** (2025).The Application of Serious Games in Inclusive Education. In *Game Industry Blue Book: China Game Industry Development Report (2025)*. Beijing: Social Sciences Academic Press [BC].
- Wei, Y., **Cheng, X.**, Shi, R., Liang, H.N. (2026).Reorienting with the Bare Hand: Gesture-Based Techniques for Orientation-Enabled Teleportation in Virtual Reality. In *IEEE transactions on visualization and computer graphics (TVCG)* [J, SCI Q1] & IEEE virtual reality (VR) [C, CCF-A].
- Gao, Y., **Cheng, X.**, Li Y., Xu, D., Sun, D., Hu, Y. ISDG-Net: Efficient RGB-Infrared Object Detection for Remote Sensing Imagery. In *Remote Sensing* [J, SCI Q1].
- **Cheng, X.**, Zhang, Y., Qiao, Y., Hansen, P., Rui, J. (2026). Negative Danmaku as Social Warning Cues in Video Credibility Assessment. In *European Conference on Computer-Supported Cooperative Work (ECSCW 2026), Posters* [C, CCF-B] (To Appear).
- Zhang, Y., **Cheng, X.**, Zhao, P., Jiang, R. Exploring the Efficacy of Virtual Reality in Enhancing Human Spatial-Visual Abilities: A Pilot Study on the Spatial Vision VR Training System. In *Games for Health Journal* [J] (Under Third Review).
- **Cheng, X.**, Zhang, Y., Wei, Y., Rui, J., Hansen, P., Liang, H.N. Effects of Spatial Constraint on Bare-Hand Logographic Character Handwriting Performance in Virtual Reality. In *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* [C] (Under Review).
- Wang, J., **Cheng, X.**, Hou, B., Liang, H.N. XR-Read: Eye-Level Clarity Matters for Sustained Reading in Head-Mounted Displays. In *SIGGRAPH Asia 2026, Technical Papers* [C] (Under Review). [arXiv: 2601.02829]

RESEARCH EXPERIENCE

Research Intern, Department of Computer Science and Engineering, Hong Kong University of Science and Technology Dec 2025 – Sep 2026

- Supervisor: Xiaojuan Ma
- Participated in the conceptualization and research design of specific tasks such as the human-computer interaction research direction related to speculative artificial products, human-machine collaboration, and the support of large language models.

Research Assistant, Computational Media and Arts Thrust, Hong Kong University of Science and Technology (Guangzhou) Jul 2025 – Sep 2026

- Supervisor: Hai-Ning Liang
- Finished experimental design, development, and user studies for three parallel projects on VR resolution, reorientation techniques, and handwriting interaction; contributed to three manuscripts accepted/submitted to IEEE TVCG/VR/ISMAR.

Visiting Student, Department of Information Management, Peking University Feb 2025 – Jun 2025

- Supervisor: Pengyi Zhang
- Led interaction design and front-end development for two AI agent-based educational systems supporting collaborative lesson planning, interactive learning, and knowledge construction; contributed to an NSFC-funded research project and an in-progress patent application.

Research Intern, National Virtual Simulation Experimental Teaching Center for Earth Sciences, Peking University Dec 2024 – Jun 2025

- Supervisor: Yanjun Guo
- Conducted research on immersive virtual environments for marine science education, investigating how sense of presence and natural connectedness influence learners' pro-environmental behavioral intentions.

Research Intern, Future Lab - Space and Media Group, Tsinghua University May 2023 – Jul 2023

- Supervisor: Keyang Tang
- Assisted in the development and design of the Digital Luoyang human-computer interaction system.

INDUSTRIAL EXPERIENCE

Investment Intern, Alpha Startups Aug 2025 – Nov 2025

- Conducted pre-investment research, focusing on next-generation interaction technologies and products.
- Participated in all stages of the investment process, including company analysis and market research.

Research Intern, MiraclePlus (YC China) Oct 2024 – Nov 2024

- Responsible for communicating with entrepreneurs and planning Demo Day events.

Product Manager Intern, Kuaishou Technology Apr 2024 – Aug 2024

- Worked on product development for the e-commerce C-end product department.

Product Manager Intern, DiDi Global Nov 2023 – Apr 2024

- Worked on growth product initiatives for the Didi platform, Chitu, and Didi Alliance.

Interaction Design Intern, Meituan Jul 2023 – Nov 2023

- Responsible for interaction design and AI capability building in Meituan Cuisine.
- Received a full-time offer after internship.

ACADEMIC SERVICES

- Teaching Assistant - Game Development Game Engines, Peking University Sep 2024 – Jan 2025
- Teaching Assistant - Artificial Intelligence Practice, Peking University Mar 2025 – Jul 2025
- Reviewer - ISMAR 2025/2026
- Reviewer - CHI 2026
- Reviewer - C&C 2026
- Reviewer - UIST 2026

HONORS & AWARDS

- **National Scholarship** (Top 1%), 2022
- People's Scholarship (Top 5%), 2021, 2022, 2023
- Shenzhen Stock Exchange Scholarship (8000 RMB, awarded to 17 students university-wide), 2025
- Outstanding Student Award, 2025
- Hollywood Independent Film Festival, Best Student Short Film, 2022
- China Functional Game Design Competition, Excellent Award, 2022
- Japanese Conceptual Art and Design Award, Third Prize, 2022
- The 4th Hong Kong Contemporary Design Award, 2023
- The 14th National College Student Advertising Art Competition, Third Prize, 2022
- The 4th 2024-GCCROSS Creative Award, Bronze Award, 2023
- AADC North American Applied Art and Design Competition, Bronze Award, 2023

SKILLS

- **Programming & Prototyping:** Python, C#, C++, Blueprint, JavaScript, HTML, CSS
- **Software:** Unity, Unreal Engine, Processing, TouchDesigner, 3DSMAX, Adobe (PS, AE, AN, PR, AI), Blender, Figma
- **Devices:** Kinect, LeapMotion, Arduino, VR/AR/MR (own Meta Quest 3, Pico 4, HoloLens 2, Insta360, HTC VIVE)
- **Statistical Methods:** Descriptive Statistics, Hypothesis Testing (t-test, Chi-Square Test, ANOVA), Regression Analysis (Linear Regression, Logistic Regression), Non-Parametric Tests
- **Statistical Software:** SPSS, Python (pandas, statsmodels), Excel, C
- **Hobbies:** Pilates, Scuba Diving, Free Diving, Mahjong, Swimming, FPS Games, Badminton, Music, Movies, Piano (Grade 10 of Wuhan Conservatory of Music), Cooking